

DecodeLabs Internship Cyber Security – Project 3

PhishGuard Phishing Awareness & Threat Analysis Tool

Project Report

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Project: 3 – Phishing Awareness Analysis

1. Executive Summary

This report presents PhishGuard, a phishing awareness and threat analysis tool developed as Project 3 of the DecodeLabs Cyber Security Internship. The project focuses on analyzing sample emails and messages to identify indicators commonly associated with phishing attempts. The DecodeLabs project brief defines the goal as analyzing sample communications to identify phishing attempts, with key requirements to identify suspicious links or keywords, list phishing red flags, and explain why a message is unsafe.

PhishGuard implements these requirements through a rule-based analysis approach. It evaluates suspicious keywords, urgency and pressure tactics, credential-related language, financial indicators, threatening language, potentially dangerous attachments, email addresses, URLs, and URL characteristics. The tool combines detected indicators into a risk score from 0 to 100 and assigns a LOW, MEDIUM, HIGH, or CRITICAL risk level. It also provides a red-flag checklist, explanations, and recommended actions.

2. Introduction

Phishing is a social-engineering technique in which attackers attempt to deceive users into revealing sensitive information, opening malicious content, or following fraudulent instructions. Effective phishing awareness therefore requires more than recognizing a single suspicious word or link; it requires considering multiple signals in the context of the message.

The DecodeLabs Project 3 brief describes this stage as a detection phase focused on threat identification and emphasizes analyzing communication for deceptive tactics and malicious intent. The required outcomes are identification of suspicious links or keywords, listing red flags, and explaining why a message is unsafe.

3. Project Objectives

- Analyze sample emails and messages for phishing-related indicators.
- Identify suspicious keywords and links.
- Detect common phishing red flags such as urgency, credential requests, threats, and suspicious attachments.
- Analyze URLs for potentially suspicious characteristics.
- Generate an interpretable risk score and risk level.
- Explain why a message may be unsafe.
- Provide recommended security actions to help users respond safely.
- Demonstrate the solution using both suspicious and legitimate test messages.

4. Requirements Mapping

DecodeLabs Requirement	PhishGuard Implementation	Status
Identify suspicious links or keywords	Keyword detection and URL extraction/analysis.	Completed
List red flags found in phishing messages	Phishing Red Flags section and Red Flag Checklist.	Completed
Explain why the message is unsafe	Why Could This Message Be Unsafe? section and	Completed

	recommended action.	
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5. System Overview

PhishGuard is implemented as a Python-based Streamlit web application. The user pastes an email or message into the analysis interface and selects the Analyze Message option. The application passes the message to the analysis component, collects detected indicators, calculates the risk assessment, and presents the results through a structured dashboard.

Main components:

- app.py – Streamlit user interface and presentation of analysis results.
- analyzer.py – Rule-based message analysis and indicator detection.
- requirements.txt – Python package dependencies for the project.
- samples/ – Sample messages used for testing.
- screenshots/ – Evidence captured from application testing.
- README.md – Project setup, features, testing information, and usage instructions.

6. Technologies Used

Technology	Purpose
Python	Core programming language.
Streamlit	Web-based interface for interacting with the analyzer.
Regular expression and text analysis logic	Detection and extraction of message indicators.
URL parsing and rule-based checks	Analysis of URLs and suspicious characteristics.
Virtual environment	Isolated Python project environment.

7. Implemented Features

Suspicious Keyword Detection: The analyzer identifies terms and phrases associated with phishing, including examples such as urgent, immediately, verify, suspended, password, login, and unusual activity.

Urgency Detection: The system detects pressure tactics such as urgent instructions, immediate action requirements, and short deadlines such as 'within 24 hours'.

Credential Indicator Detection: The tool identifies language related to usernames, passwords, login information, account verification, and other credential-related requests.

Financial Indicator Detection: The analyzer checks for financial-related language such as payment and billing requests.

Threat Indicator Detection: Threatening or intimidating consequences such as account suspension or failure-to-comply statements are identified.

Attachment Analysis: Potentially dangerous attachment extensions are flagged, including executable-style files such as .exe.

Email Address Extraction: Email addresses found in the submitted message are displayed separately for inspection.

URL Analysis: URLs are extracted and analyzed for characteristics such as use of HTTP instead of HTTPS and use of an IP address rather than a domain name.

Risk Scoring: Detected indicators contribute to an overall score from 0 to 100 and a corresponding risk level.

Red Flag Checklist: The dashboard provides a consolidated checklist showing which phishing indicators were detected.

Safety Explanation: The application explains why the message could be unsafe and provides recommended security actions.

8. Risk Assessment Model

PhishGuard presents the final assessment as a score between 0 and 100. The score is interpreted using four risk categories:

Score	Risk Level	Interpretation
0–24	LOW	No significant phishing indicators detected by the rule set.
25–49	MEDIUM	Some suspicious characteristics detected; additional verification recommended.
50–74	HIGH	Several suspicious characteristics detected; careful verification required.
75–100	CRITICAL	Multiple strong phishing indicators detected; treat the message as highly suspicious.

9. Testing and Validation

The completed application was tested with six sample messages covering both phishing-style and legitimate communications. The test set included urgent account suspension content, a fake payment request, a suspicious attachment, a URL shortener, a legitimate university library notice, and a legitimate student-services notification.

Test Case	Scenario	Expected Classification	Observed Result
01	Urgent account suspension	HIGH / CRITICAL	CRITICAL – 82/100
02	Fake payment request	HIGH / CRITICAL	Phishing indicators detected
03	Suspicious executable attachment	HIGH / CRITICAL	Phishing indicators detected
04	URL shortener and credential request	HIGH / CRITICAL	Phishing indicators detected
05	Legitimate university library message	LOW	LOW – 0/100
06	Legitimate student-services notification	LOW	LOW

10. Detailed Test Result: Critical Phishing Example

One of the main validation tests used an account-security message containing an urgent deadline, an account suspension threat, a credential request, an HTTP URL using a private IP address, and a .exe attachment. The analyzer produced a risk score of 82/100 and classified the message as CRITICAL.

Detected indicators included:

- Urgency: urgent, immediately, within 24 hours.
- Credential indicators: password, username, login, verify your account.
- Threat indicators: account suspension and failure to comply.
- Potentially dangerous attachment: .exe.
- URL findings: HTTP rather than HTTPS and an IP address used instead of a domain.
- Multiple phishing red flags and suspicious keywords.

11. Detailed Test Result: Legitimate Example

A legitimate university-style message containing routine library information and no suspicious request was also tested. The analyzer produced a score of 0/100 and classified the message as LOW. No suspicious keywords, urgency indicators, credential indicators, URL findings, financial indicators, threats, or dangerous attachments were detected.

12. Security Recommendations Produced by the Tool

- Do not click suspicious links.
- Do not open unexpected or suspicious attachments.
- Do not provide passwords or other sensitive information in response to suspicious messages.
- Verify unexpected requests through an official communication channel.
- Review the sender, URL, wording, urgency, and requested action before responding.

13. Limitations

- The analyzer is rule-based and therefore depends on the indicators and patterns defined in the application.
- A sophisticated phishing message that avoids the configured indicators may not receive a high risk score.
- A legitimate message containing suspicious-looking words could potentially be flagged.
- Risk scoring should be treated as an awareness aid rather than definitive proof that a message is malicious or safe.
- The project is designed for educational and awareness purposes and is not presented as an enterprise email-security gateway.

14. Future Improvements

- Expand the indicator and keyword database.
- Add domain reputation and URL reputation checking through trusted security services.
- Improve detection of URL shorteners and redirect chains.
- Add sender-domain and display-name analysis.

- Support file metadata and safer attachment inspection.
- Introduce a larger labelled dataset for systematic evaluation.
- Add machine-learning-based classification as a future enhancement while retaining explainable rule-based indicators.

15. Skills Demonstrated

- Threat identification and phishing awareness.
- Rule-based security analysis.
- Text processing and pattern detection.
- URL security analysis.
- Risk assessment and security decision-making.
- Python application development.
- Streamlit user-interface development.
- Testing using both malicious-looking and legitimate sample messages.
- Cybersecurity documentation and portfolio preparation.

16. Conclusion

PhishGuard successfully implements the core objectives of DecodeLabs Cyber Security Project 3. The application analyzes sample emails and messages, identifies suspicious links and keywords, lists phishing red flags, and explains why potentially dangerous messages should be treated cautiously. The addition of a Red Flag Checklist also supports the project brief's emphasis on practical threat-identification and employee awareness.

Testing demonstrated that the application can distinguish between a clean legitimate message and a message containing multiple phishing indicators. In the primary phishing test, the system produced a CRITICAL score of 82/100, while the legitimate university test produced a LOW score of 0/100. These results provide practical evidence that the implemented rule-based approach is functioning as intended for the selected test cases.

17. Reference

DecodeLabs, "Cyber Security Project 3 – Phishing Awareness Analysis," Industrial Training Kit, Batch 2026. The project brief defines the project goal and key requirements used as the basis for this report.

Appendix A – Suggested Evidence Screenshots

- Screenshot 1 – CRITICAL phishing analysis showing risk score, detections, URL analysis, red flags, and checklist.
- Screenshot 2 – URL shortener analysis showing URL-related indicators.
- Screenshot 3 – Legitimate message analysis showing LOW risk and no major indicators.

Appendix B – Project Folder Structure

Project-3-Phishing-Awareness/
├── app.py

```
|— analyzer.py
|— requirements.txt
|— README.md
|— .gitignore
|— samples/
|   |— 01_urgent_account.txt
|   |— 02_fake_payment.txt
|   |— 03_suspicious_attachment.txt
|   |— 04_url_shortener.txt
|   |— 05_legitimate_university.txt
|   |— 06_legitimate_notification.txt
|— screenshots/
|— report/
```